

Quadratic Formula

Equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Proof

Starting from the quadratic equation with coefficients a, b, c and $a \neq 0$,

$$ax^2 + bx + c = 0$$

Dividing both sides by a

$$x^2 + \frac{bx}{a} + \frac{c}{a} = 0$$

Subtracting $\frac{c}{a}$,

$$x^2 + \frac{bx}{a} = -\frac{c}{a}$$

Now, we can complete the square: observe that the first term can be x and the second can be $\frac{b}{2a}$, so that $2 \cdot x \cdot \frac{b}{2a}$ becomes $\frac{bx}{a}$. To do this, we add the square of $\frac{b}{2a}$ in both sides, obtaining

$$x^2 + \frac{b}{a} \cdot x + \frac{b^2}{4a^2} = \frac{b^2}{4a^2} - \frac{c}{a} \quad (\text{I})$$

Note that the left side of the equation is a perfect square of the form $a^2 + 2ab + b^2 = (a + b)^2$, so,

$$x^2 + 2 \cdot \frac{b}{2a} \cdot x + \left(\frac{b}{2a}\right)^2 = \left(x + \frac{b}{2a}\right)^2$$

And the right side can be adjusted to the same denominator by the multiplication of $\frac{c}{a}$ by $\frac{4a}{4a}$

$$\frac{b^2}{4a^2} - \frac{c}{a} \cdot \frac{4a}{4a} = \frac{b^2 - 4ac}{4a^2}$$

Returning to equation I,

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Taking the square root of both sides,

$$\sqrt{\left(x + \frac{b}{2a}\right)^2} = \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

$$\left|x + \frac{b}{2a}\right| = \frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

Finally, subtracting $\frac{b}{2a}$ from both sides, we have

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$